

Jungho Lee

PH.D CANDIDATE

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RESEARCH INTERESTS

3D Vision

- World Model
- Feedforward Gaussian Splatting
- 3D Scene Reconstruction

EDUCATION

Yonsei University | College of Engineering

Seoul, South Korea

INTERATED M.S./PH.D IN ELECTRICAL AND ELECTRONIC ENGINEERING

Sep. 2021 - Aug. 2026 (Expected)

- Image and Video Pattern Recognition Lab.
- Advisor: Prof. Sangyoun Lee

Yonsei University | College of Engineering

Seoul, South Korea

B.S. IN ELECTRICAL & ELECTRONIC ENGINEERING

Mar. 2015 - Aug. 2021

- 2-Year Military Service (2017-2019)

RESEARCH EXPERIENCE

NAVER Cloud

Seongnam, South Korea

RESEARCH INTERN

Aug. 2024 - Feb. 2025

- 3D Scene Representation from Defocused Images
- 3D Human Avater Generation
- Mentor: Ho-Deok Jang

PUBLICATIONS

First-Author Papers

SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes

2026

JUNGHOO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE

Findings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)

CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images

2025

JUNGHOO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOON WEE, SANGYOUN LEE

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)

CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images

2025

JUNGHOO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE

Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)

SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields

2025

JUNGHOO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE

2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)

Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition

2023

JUNGHOO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)

Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition

2023

JUNGHOO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)

Conference Proceedings

- MoRGs: Efficient Per-Gaussian Motion Reasoning for Streamable Dynamic 3D Scenes** [2026](#)
WONJOON LEE, SUNGMIN WOO, DONGHYEONG KIM, **JUNGHO LEE**, SANGHEON PARK, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- SwiftVGGT: A Scalable Visual Geometry Grounded Transformer for Large-Scale Scenes** [2026](#)
JUNGHO LEE, MINHYEOK LEE, SUNGHUN YANG, MINSEOK KANG, SANGYOUN LEE
Findings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- MonoCLUE: Object-Aware Clustering Enhances Monocular 3D Object Detection** [2026](#)
SUNGHUN YANG, MINHYEOK LEE, **JUNGHO LEE**, SANGYOUN LEE
The Association for the Advancement of Artificial Intelligence (AAAI)
- GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection** [2025](#)
DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, **JUNGHO LEE**, SANGYOUN LEE
The 3rd Workshop in Vision-based Industrial InspectiON (ICCVW)
- Find First, Track Next: Decoupling Identification and Propagation in Referring Video Object Segmentation** [2025](#)
SUHWAN CHO, SEUNGHOO LEE, MINHYEOK LEE, **JUNGHO LEE**, SANGYOUN LEE
The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)
- DepthFlow: Exploiting Depth-Flow Structural Correlations for Unsupervised Video Object Segmentation** [2025](#)
SUHWAN CHO, MINHYEOK LEE, **JUNGHO LEE**, DONGHYEONG KIM, SANGYOUN LEE
The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)
- TransFlow: Motion Knowledge Transfer from Video Diffusion Models to Video Salient Object Detection** [2025](#)
SUHWAN CHO, MINHYEOK LEE, **JUNGHO LEE**, SUNGHUN YANG, SANGYOUN LEE
The 7th Large-Scale Object Segmentation (LSVOS) Workshop (ICCVW)
- CoMoGaussian: Continuous Motion-Aware Gaussian Splatting from Motion-Blurred Images** [2025](#)
JUNGHO LEE, DONGHYEONG KIM, DOGYOON LEE, SUHWAN CHO, MINHYEOK LEE, WONJOON LEE, TAEHO KIM, DONGYOON WEE, SANGYOUN LEE
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)
- CMTM: Cross-Modal Token Modulation for Unsupervised Video Object Segmentation** [2025](#)
INSEOK JEON, SUHWAN CHO, MINHYEOK LEE, SEUNGHOO LEE, MINSEOK KANG, **JUNGHO LEE**, CHAEWON PARK, DONGHYEONG KIM, SANGYOUN LEE
IEEE International Conference on Image Processing (ICIP)
- CoCoGaussian: Leveraging Circle of Confusion for Gaussian Splatting from Defocused Images** [2025](#)
JUNGHO LEE, SUHWAN CHO, TAEHO KIM, HO-DEOK JANG, MINHYEOK LEE, GEONHO CHA, DONGYOON WEE, DOGYOON LEE, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- Effective SAM Combination for Open-Vocabulary Semantic Segmentation** [2025](#)
MINHYEOK LEE, SUHWAN CHO, **JUNGHO LEE**, SUNGHUN YANG, HEESEUNG CHOI, IG-JAE KIM, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Oral Presentation
- SMURF: Continuous Dynamics for Motion-Deblurring Radiance Fields** [2025](#)
JUNGHO LEE, DOGYOON LEE, MINHYEOK LEE, DONGHYEONG KIM, SANGYOUN LEE
2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPRW)
- Video Diffusion Models are Strong Video Inpainter** [2025](#)
MINHYEOK LEE, SUHWAN CHO, CHAJIN SHIN, **JUNGHO LEE**, SUNGHUN YANG, SANGYOUN LEE
The Association for the Advancement of Artificial Intelligence (AAAI)
- Guided Slot Attention for Unsupervised Video Object Segmentation** [2024](#)
MINHYEOK LEE, DOGYOON LEE, SUHWAN CHO, CHAEWON PARK, **JUNGHO LEE**, SANGYOUN LEE
Proceedings of the IEEE/CVF Computer Vision and Pattern Recognition (CVPR)
- Hierarchically Decomposed Graph Convolutional Networks for Skeleton-Based Action Recognition** [2023](#)
JUNGHO LEE, MINHYEOK LEE, DOGYOON LEE, SANGYOUN LEE
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)

- Leveraging Spatio-Temporal Dependency for Skeleton-Based Action Recognition** 2023
JUNGHO LEE, MINHYEOK LEE, SUHWAN CHO, SUNGMIN WOO, SUNGJUN JANG, SANGYOUN LEE
Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)
- Detection-Identification Balancing Margin Loss for One-Stage Multi-Object Tracking** 2022
HEANSUNG LEE, SUHWAN CHO, SUNGJUN JANG, JUNGHO LEE, SANGYOUN LEE
IEEE International Conference on Image Processing (ICIP)

Journals

- GenCLIP: Generalizing CLIP Prompts for Zero-Shot Anomaly Detection** 2025
DONGHYEONG KIM, CHAEWON PARK, SUHWAN CHO, HYEONJEONG LIM, MINSEOK KANG, JUNGHO LEE, SANGYOUN LEE
Pattern Recognition (PR)
- Sparse-DeRF: Deblurred Neural Radiance Fields from Sparse View** 2025
DOGYOON LEE, DONGHYEONG KIM, JUNGHO LEE, MINHYEOK LEE, SEUNGHOON LEE, SANGYOUN LEE
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Treating Motion as Option with Output Selection for Unsupervised Video Object Segmentation** 2025
SUHWAN CHO, MINHYEOK LEE, JUNGHO LEE, MYEONGAH CHO, SANGYOUN LEE
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Multi-Scale Structural Graph Convolutional Network for Skeleton-Based Action Recognition** 2024
SUNGJUN JANG, HEANSUNG LEE, WOJIN KIM, JUNGHO LEE, SUNGMIN WOO, SANGYOUN LEE
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

Preprinted Papers

- [Submitted to ECCV] M. Kang, M. Lee, J. Lee, D. Kim, S. Woo, M. Kim, I. Jeon, S. Lee. Revisiting Weakly-Supervised Video Scene Graph Generation via Pair Affinity Learning.
- [Submitted to TPAMI] M. Lee, J. Lee, S. Cho, S. Yang, S. Lee. ESC-Net++: Effective SAM Combination for Open-Vocabulary Image and Video Segmentation.
- [Submitted to PR] M. Lee, D. Lee, J. Lee, S. Cho, S. Lee. Synchronizing Vision and Language: Bidirectional Token-Masking AutoEncoder for Referring Image Segmentation. *arXiv preprint arXiv:2311.17952*, 2025.

PATENTS

Domestic Patent

- [P3] Palmprint Recognition Method. KR-Application No.10-2023-0156996, Nov., 2023.
- [P2] Skeleton Graph-based Action Recognition Device and Method. KR-Application No.10-2023-0123693, Sep., 2023.
- [P1] Video Anomaly Detection Apparatus and Method using Relational Embedding. KR-Application No.10-2022-0156968, Nov., 2022.

PROJECTS

- Research on Robust Neural Rendering-Based Large-Scale 3D Ultra-Precision Virtual Space Generation and Spatial Registration for Low-Quality Noisy Data** *National Research Foundation of Korea*
DEEP LEARNING RESEARCHER *May. 2024 - Apr. 2027*
- Development of large scene reconstruction by 3D Gaussian Splatting.
- Collaborative Perception and Intelligence Framework for Hyper-connected Interaction among Human and Intelligent Things** *Korea Electronics Technology Institute*
DEEP LEARNING RESEARCHER *Apr. 2024 - Dec. 2025*
- Development of efficient skeleton-based action recognition model.
- Development of Anti-spoofing Model for Face Recognition Based on RGB Camera** *Samsung Electronics*
DEEP LEARNING RESEARCHER *Aug. 2023 - Jul. 2024*
- Development of face anti-spoofing model robust to various spoofing attack.
- Development of Mobile Palmprint Recognition Algorithm** *Samsung Electronics*
DEEP LEARNING RESEARCHER *Aug. 2022 - Jul. 2023*
- Development of one-stage real-time mobile network, which includes keypoint detection and palmprint recognition.
 - Development of real-time Android demo application for palmprint recognition.

Deep Learning-Based Initial Identification and Tracking System for Missing Persons in Heterogeneous CCTV Images

DEEP LEARNING RESEARCHER

- Development of real-time multi-object tracking algorithm robust to occluded person.

National Research Foundation of
Korea

Oct. 2018 - Dec. 2022

Development of AI Multi-Object Tracking and Behavior Analysis Technology

DEEP LEARNING RESEARCHER

- Development of robust feature extractor for the object detection network.

Hanwha Techwin

Oct. 2020 - Oct. 2021

Professional Services

Invited Talks

- "3D Neural View Synthesis from Degraded Images," NAVER AI Lab Jul. 2025
- "3D Neural View Synthesis from Degraded Images," Korean Electronics Technology Institute Jul. 2025
- "Leveraging Circle of Confusion for 3D Neural View Synthesis," Korean Photonics Technology Institute May. 2025

Journal Reviewer

- IEEE Transactions on Graphics (TOG) 2025
- International Journal of Computer Vision (IJCV) 2024
- IEEE Transactions on Multimedia (TMM) 2024
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 2023
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2023
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) 2023

Conference Reviewer

- Association for the Advancement of Artificial Intelligence (AAAI) 2026
- IEEE/CVF Computer Vision and Pattern Recognition (CVPR) - Outstanding Reviewer (top 5%) 2025
- Conference on Neural Information Processing Systems (NeurIPS) 2024
- European Conference on Computer Vision (ECCV) 2024
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024

TEACHING EXPERIENCES

Understanding and Using AI

TEACHING ASSISTANT

Yonsei University
Spring 2022 - Spring 2025

Deep Learning Lab.

TEACHING ASSISTANT

Yonsei University
Spring 2023

Digital Logic Circuit

TEACHING ASSISTANT

Yonsei University
Fall 2021

SKILLS

Research and Development Stacks

- Main Languages** Python, C/C++, MATLAB, Kotlin
- Machine Learning** PyTorch, TensorFlow, Keras
- Computer Vision** OpenCV